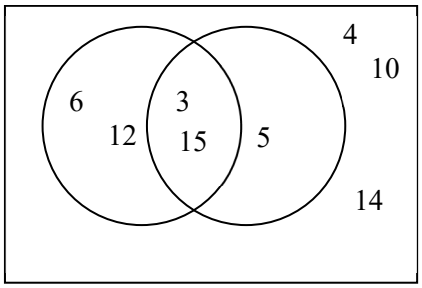
Abbreviations

cao	correct answer only
dep	dependent
FT	follow through after error
isw	ignore subsequent working
oe	or equivalent
SC	Special Case
nfww	not from wrong working
soi	seen or implied

Question	Answer	Marks	Partial Marks
1(a)	Kite	**1**	
1(b)	4	**1**	
1(c)(i)	Enlargement [centre] (−8, 2) [scale factor] 2	**3**	**B1** for each
1(c)(ii)	Translation $\begin{pmatrix} 0 \\ -10 \end{pmatrix}$ OR Reflection in $y = -1$ oe	**2**	**B1** for each
1(d)(i)	Correct rotation (6, −1), (7, −4), (6, −5), (5, −4)	**2**	**B1** for a correct 90° clockwise rotation about (0, 0) or correct orientation, incorrect position
1(d)(ii)	Correct reflection (3, −6), (6, −5), (7, −6), (6, −7)	**2**	**B1** for a correct reflection in $y = 1$ or in $x = k$, $k \neq 1$
2(a)	$A \cup B$	**1**	
2(b)(i)	1 4 9 16	**2**	**B1** for 3 correct and none incorrect or for all correct and one extra incorrect
2(b)(ii)	2, 3, 5, 7, 11, 13	**2**	**B1** for 5 correct and none or one incorrect
2(b)(iii)(a)		**2**	**B1** for 2 or 3 regions correct
2(b)(iii)(b)	7 11 13	**1**	**FT** *their* Venn diagram
2(b)(iii)(c)	2	**1**	**FT** *their* Venn diagram

Question	Answer	Marks	Partial Marks
2(b)(iii)(d)	$\frac{4}{16}$ oe	**1**	**FT** *their* Venn diagram for $\frac{n(F)}{16}$
3(a)	$2 \times 630 + 3 \times \frac{5}{8} \times 630$	**M2**	**M1** for $[3 \times] \frac{5}{8} \times 630$
	2441.25	**A1**	
3(b)(i)	13 [h] 50 [min]	**3**	**B2** for 16[h] 20[min] or 18[h] 50[min] as final answer or 13[h] 50[min] not as final answer or 16[h] 20[min] with – 2[h] 30[min] attempted or 18[h] 50[min] with – 5[h] attempted or **B1** for 21[h] 20[min] seen or 16[h] 20[min] or 18[h]50 [min] seen or 23[h] 50[min] as final answer or an attempt made to find a time of flight
3(b)(ii)	13.833 cao	**1**	**FT** *their* time correct to 3 decimal places
3(b)(iii)	10 038	**1**	
3(b)(iv)	726 or 725.6 to 725.7	**2**	**FT** *their* **(b)(iii)** ÷ *their* **(b)(ii)** correctly evaluated **M1** for *their* **(b)(iii)** ÷ *their* **(b)(ii)**
3(c)	2400 ÷ 8 × (8 + 5 + 6) [=5700]	**M2**	**M1** for 2400 ÷ 8
4(a)(i)	Octagon	**1**	
4(a)(ii)	180 – (360 ÷ 8) or $\frac{(8-2)\times 180}{8}$	**M2**	**M1** for 360 ÷ 8 or (8 – 2) × 180
4(b)(i)	22 29 36	**3**	**B1** for each or **B2FT** for adding 7 twice or **B1FT** for adding 7 between terms once
4(b)(ii)	Add 7 oe	**1**	
4(b)(iii)	64	**1**	
4(b)(iv)	$7n + 1$ oe final answer	**2**	**M1** for $jn + 1, j \neq 0$ or $7n + k, k \neq 1$ or for $7n + 1$ oe seen but not as final answer
4(b)(v)	16 nfww	**2**	**M1** for *their* **(b)(iv)** = 113
5(a)	49	**1**	
5(b)	9002	**1**	